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Methods for evaluating the integrity of a uterine cavity

Abstract

Methods, systems and devices for evaluating the integrity of a uterine cavity. A method comprises introducing transcervically a probe into a patient's uterine cavity, providing a flow of a fluid (e.g., CO.sub.2) through the probe into the uterine cavity and monitoring the rate of the flow to characterize the uterine cavity as perforated or non-perforated based on a change in the flow rate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/532,801, filed Nov. 22, 2021, now U.S. Pat. No. 11,832,875, which is a continuation of U.S. patent application Ser. No. 15/844,050, filed Dec. 15, 2017, now U.S. Pat. No. 11,191,588, which is a continuation of U.S. patent application Ser. No. 13/488,214, filed Jun. 4, 2012, now U.S. Pat. No. 9,883,907, which is a continuation of U.S. patent application Ser. No. 13/442,449, filed Apr. 9, 2012, now U.S. Pat. No. 9,788,890, which claims the benefit of U.S. Provisional Application No. 61/483,542, filed May 6, 2011; and U.S. Provisional Application No. 61/491,842, filed May 31, 2011, the entire content of each of which are fully incorporated herein by reference.

BACKGROUND

1. Field of the Invention

(1) The present invention relates to electrosurgical methods and devices for global endometrial ablation in a treatment of menorrhagia. More particularly, the present invention relates to applying radiofrequency current to endometrial tissue by means of capacitively coupling the current through an expandable, thin-wall dielectric member enclosing an ionized gas.

(2) A variety of devices have been developed or proposed for endometrial ablation. Of relevance to the present invention, a variety of radiofrequency ablation devices have been proposed including solid electrodes, balloon electrodes, metalized fabric electrodes, and the like. While often effective,

many of the prior electrode designs have suffered from one or more deficiencies, such as relatively slow treatment times, incomplete treatments, non-uniform ablation depths, and risk of injury to adjacent organs.

(3) For these reasons, it would be desirable to provide systems and methods that allow for endometrial ablation using radiofrequency current which is rapid, provides for controlled ablation depth and which reduce the risk of injury to adjacent organs. At least some of these objectives will be met by the invention described herein.

2. Description of the Background Art

(4) U.S. Pat. Nos. 5,769,880; 6,296,639; 6,663,626; and 6,813,520 describe intrauterine ablation devices formed from a permeable mesh defining electrodes for the application of radiofrequency energy to ablate uterine tissue. U.S. Pat. No. 4,979,948 describes a balloon filled with an electrolyte solution for applying radiofrequency current to a mucosal layer via capacitive coupling. US 2008/097425, having common inventorship with the present application, describes delivering a pressurized flow of a liquid medium which carries a radiofrequency current to tissue, where the liquid is ignited into a plasma as it passes through flow orifices. U.S. Pat. No. 5,891,134 describes a radiofrequency heater within an enclosed balloon. U.S. Pat. No. 6,041,260 describes radiofrequency electrodes distributed over the exterior surface of a balloon which is inflated in a body cavity to be treated. U.S. Pat. No. 7,371,231 and US 2009/054892 describe a conductive balloon having an exterior surface which acts as an electrode for performing endometrial ablation. U.S. Pat. No. 5,191,883 describes bipolar heating of a medium within a balloon for thermal ablation. U.S. Pat. Nos. 6,736,811 and 5,925,038 show an inflatable conductive electrode.

BRIEF SUMMARY

(5) The following presents a simplified summary of some embodiments of the invention in order to provide a basic understanding of the invention. This summary is not an extensive overview of the invention. It is not intended to identify key/critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some embodiments of the invention in a simplified form as a prelude to the more detailed description that is presented later.

(6) The present invention provides methods, systems and devices for evaluating the integrity of a uterine cavity. The uterine cavity may be perforated or otherwise damaged by the transcervical introduction of probes and instruments into the uterine cavity. If the uterine wall is perforated, it would be preferable to defer any ablation treatment until the uterine wall is healed.

(7) A method of the invention comprises introducing transcervically a probe into a patient's uterine cavity, providing a flow of a fluid (e.g., CO₂) through the probe into the uterine cavity and monitoring the rate of the flow to characterize the uterine cavity as perforated or non-perforated based on a change in the flow rate. If the flow rate into the cavity drops to zero or close to zero within a predetermined time period, this indicates that the uterine cavity is intact and not perforated. If the flow rate does not drop to zero or close to zero, this indicates that a fluid flow is leaking through a perforation in the uterine cavity into the uterine cavity or escaping around an occlusion balloon that occludes the cervical canal.

(8) Embodiments herein provide a method of characterizing a patient's uterus, comprising introducing a flow of a fluid into a uterine cavity of a patient; and monitoring the flow to characterize the uterine cavity as at least one of perforated or non-perforated based on a change in a rate of the flow. Introducing may be, for example, transcervically introducing a probe into the uterine cavity, and introducing the flow through the probe.

(9) Monitoring may include providing a signal, responsive to the rate of flow, that characterizes the uterine cavity as at least one of perforated or non-perforated. As an example, monitoring may include generating a signal responsive to the rate of flow not dropping below a predetermined level, the signal characterizing the uterine cavity as perforated. In embodiments, the predetermined level is 0.05 slpm.

(10) In embodiments, monitoring comprises generating a signal responsive to the rate of flow

dropping below a predetermined level, the signal characterizing the uterine cavity as non-perforated. The predetermined level may be, for example, 0.05 slpm.

(11) In further embodiments, monitoring comprises monitoring a rate of flow after a predetermined first interval after initiation of the flow. The first interval may be, as examples, at least 5 seconds, at least 15 seconds, or at least 30 seconds.

(12) Monitoring may additionally include monitoring a rate of flow over a second predetermined interval after the first interval. The second interval may be a least 1 second, at least 5 seconds, or at least 10 seconds, as examples.

(13) In additional embodiments, monitoring includes providing a signal, responsive to the rate of flow, that characterizes the uterine cavity as at least one of perforated or non-perforated, and wherein the signal is at least one of visual, aural and tactile.

(14) In embodiments, prior to introducing the flow, a member is positioned within the cervical canal that substantially prevents a flow of the fluid out of the uterine cavity. Introducing may include transcervically introducing a probe into the uterine cavity, and introducing the flow through the probe, with the member positioned about an exterior of the probe. The member may be expanded in the cervical canal.

(15) In embodiments, the fluid is a gas or a liquid.

(16) In additional embodiments, introducing includes transcervically introducing a probe into the uterine cavity, and introducing the flow through the probe. The probe has a working end with an energy-delivery surface for ablating uterine cavity tissue. Responsive to the uterine cavity being characterized as perforated, energy delivery surface is disabled. Alternatively or additionally, responsive to the uterine cavity being characterized as non-perforated, activation of the energy delivery surface may be enabled or even caused to happen automatically.

(17) In embodiments, a method of endometrial ablation is provided, the method including introducing an ablation probe into a uterine cavity of a patient; flowing a fluid from a fluid source through the probe into the uterine cavity; monitoring the rate of the flow of the fluid into the uterine cavity to characterize the cavity as at least one of perforated or non-perforated based on a change in the flow rate; and responsive to the uterine cavity being characterized as non-perforated, activating the ablation probe to ablate an interior of the uterine cavity.

(18) For a fuller understanding of the nature and advantages of the present invention, reference should be made to the ensuing detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to better understand the invention and to see how it may be carried out in practice, some preferred embodiments are next described, by way of non-limiting examples only, with reference to the accompanying drawings, in which like reference characters denote corresponding features consistently throughout similar embodiments in the attached drawings.

(2) FIG. 1 is a perspective view of an ablation system corresponding to the invention, including a hand-held electrosurgical device for endometrial ablation, RF power source, gas source and controller.

(3) FIG. 2 is a view of the hand-held electrosurgical device of FIG. 1 with a deployed, expanded thin-wall dielectric structure.

(4) FIG. 3 is a block diagram of components of one electrosurgical system corresponding to the invention.

(5) FIG. 4 is a block diagram of the gas flow components of the electrosurgical system of FIG. 1.

(6) FIG. 5 is an enlarged perspective view of the expanded thin-wall dielectric structure, showing an expandable-collapsible frame with the thin dielectric wall in phantom view.

(7) FIG. 6 is a partial sectional view of the expanded thin-wall dielectric structure of FIG. 5 showing (i) translatable members of the expandable-collapsible frame a that move the structure between collapsed and (ii) gas inflow and outflow lumens.

(8) FIG. 7 is a sectional view of an introducer sleeve showing various lumens of the introducer sleeve taken along line 7-7 of FIG. 6.

(9) FIG. 8A is an enlarged schematic view of an aspect of a method of the invention illustrating the step introducing an introducer sleeve into a patient's uterus.

(10) FIG. 8B is a schematic view of a subsequent step of retracting the introducer sleeve to expose a collapsed thin-wall dielectric structure and internal frame in the uterine cavity.

(11) FIG. 8C is a schematic view of subsequent steps of the method, including, (i) actuating the internal frame to move the a collapsed thin-wall dielectric structure to an expanded configuration, (ii) inflating a cervical-sealing balloon carried on the introducer sleeve, and (iii) actuating gas flows and applying RF energy to contemporaneously ionize the gas in the interior chamber and cause capacitive coupling of current through the thin-wall dielectric structure to cause ohmic heating in the engaged tissue indicated by current flow paths.

(12) FIG. 8D is a schematic view of a subsequent steps of the method, including: (i) advancing the introducer sleeve over the thin-wall dielectric structure to collapse it into an interior bore shown in phantom view, and (ii) withdrawing the introducer sleeve and dielectric structure from the uterine cavity.

(13) FIG. 9 is a cut-away perspective view of an alternative expanded thin-wall dielectric structure similar to that of FIGS. 5 and 6 show an alternative electrode configuration.

(14) FIG. 10 is an enlarged cut-away view of a portion of the expanded thin-wall dielectric structure of FIG. 9 showing the electrode configuration.

(15) FIG. 11 is a schematic view of a patient uterus depicting a method corresponding to the invention including providing a flow of a fluid media into the uterine cavity and monitoring the flow rate to characterize the patient's uterine cavity as intact and non-perforated.

(16) FIG. 12 is a perspective view of the ablation device of FIGS. 1-2 with a subsystem for checking the integrity of a uterine cavity.

(17) FIG. 13 represents a block diagram of a subsystem of the invention for providing and monitoring a fluid flow into the patient's uterine cavity.

(18) FIG. 14 represents a diagram indicating the steps of an algorithm for providing and monitoring a fluid flow into the patient's uterine cavity.

(19) FIG. 15 is a chart illustrating gas flow rates into the uterine cavity over time that will result in three conditions to thereby characterize the uterine cavity as non-perforated or perforated.

(20) FIG. 16 represents a diagram indicating the steps of an algorithm for providing and monitoring a fluid flow related to the test method of FIG. 15.

(21) FIG. 17 is a schematic view of another system and method for providing and monitoring a fluid flow to characterize the integrity of a uterine cavity.

(22) FIG. 18A is a schematic view of system with an expandable working end properly deployed in the uterine cavity, and illustrates a variation of a method for characterizing the integrity of a uterine cavity, wherein a first stage of a two-stage test which monitors CO.sub.2 flows into the uterine cavity exterior of the working end or dielectric structure.

(23) FIG. 18B illustrates the second stage of the two-stage test which monitors CO.sub.2 flows in the uterine cavity exterior of the dielectric structure in response to Argon gas flows into the interior chamber of the dielectric structure.

(24) FIG. 19 is a box diagram illustrating the steps of a second stage test as illustrated in FIG. 18B.

(25) FIG. 20A is a schematic view of an expandable working end that is not deployed in the uterine cavity and positioned in a uterine wall perforation, and illustrates a first stage of a two-stage test which monitors CO.sub.2 flows into the uterine cavity exterior of the working end.

(26) FIG. 20B illustrates the second stage of the two-stage test which monitors CO.sub.2 flows in

the uterine cavity exterior of the dielectric structure in response to Argon gas flows into the interior chamber of the dielectric structure.

(27) FIG. **21** is a box diagram illustrating the steps of a variation of the second stage of a two-stage test.

(28) FIG. **22A** is a schematic view of an expandable dielectric deployed in the uterine cavity and illustrates a test in which the dielectric is suctioned against an interior frame by a negative pressure source.

(29) FIG. **22B** is a schematic view of the expandable dielectric of FIG. **22A** improperly deployed in a perforated wall of a uterine cavity and illustrates the test wherein the dielectric is suctioned against the interior frame by the negative pressure source.

(30) FIG. **23** is an expanded schematic view of the expandable dielectric in the uterine wall perforation illustrating the escape of CO.sub.2 gas and test failure mode.

(31) FIG. **24** is a chart illustrating a combination two-stage cavity integrity test wherein the first stage comprises the test described in FIGS. **15-16** and the second stage comprises the test described in FIGS. **22A-23**.

DETAILED DESCRIPTION

(32) In the following description, various embodiments of the present invention will be described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the embodiments. However, it will also be apparent to one skilled in the art that the present invention may be practiced without the specific details. Furthermore, well-known features may be omitted or simplified in order not to obscure the embodiment being described.

(33) In general, an electrosurgical ablation system is described herein that comprises an elongated introducer member for accessing a patient's uterine cavity with a working end that deploys an expandable thin-wall dielectric structure containing an electrically non-conductive gas as a dielectric. In one embodiment, an interior chamber of the thin-wall dielectric structure contains a circulating neutral gas such as argon. An RF power source provides current that is coupled to the neutral gas flow by a first polarity electrode disposed within the interior chamber and a second polarity electrode at an exterior of the working end. The gas flow, which is converted to a conductive plasma by an electrode arrangement, functions as a switching mechanism that permits current flow to engaged endometrial tissue only when the voltage across the combination of the gas, the thin-wall dielectric structure and the engaged tissue reaches a threshold that causes capacitive coupling across the thin-wall dielectric material. By capacitively coupling current to tissue in this manner, the system provides a substantially uniform tissue effect within all tissue in contact with the expanded dielectric structure. Further, the invention allows the neutral gas to be created contemporaneously with the capacitive coupling of current to tissue.

(34) In general, this disclosure may use the terms “plasma”, “conductive gas” and “ionized gas” interchangeably. A plasma consists of a state of matter in which electrons in a neutral gas are stripped or “ionized” from their molecules or atoms. Such plasmas can be formed by application of an electric field or by high temperatures. In a neutral gas, electrical conductivity is non-existent or very low. Neutral gases act as a dielectric or insulator until the electric field reaches a breakdown value, freeing the electrons from the atoms in an avalanche process thus forming a plasma. Such a plasma provides mobile electrons and positive ions, and acts as a conductor which supports electric currents and can form spark or arc. Due to their lower mass, the electrons in a plasma accelerate more quickly in response to an electric field than the heavier positive ions, and hence carry the bulk of the current.

(35) FIG. **1** depicts one embodiment of an electrosurgical ablation system **100** configured for endometrial ablation. The system **100** includes a hand-held apparatus **105** with a proximal handle **106** shaped for grasping with a human hand that is coupled to an elongated introducer sleeve **110** having axis **111** that extends to a distal end **112**. The introducer sleeve **110** can be fabricated of a

thin-wall plastic, composite, ceramic or metal in a round or oval cross-section having a diameter or major axis ranging from about 4 mm to 8 mm in at least a distal portion of the sleeve that accesses the uterine cavity. The handle **106** is fabricated of an electrically insulative material such as a molded plastic with a pistol-grip having first and second portions, **114a** and **114b**, that can be squeezed toward one another to translate an elongated translatable sleeve **115** which is housed in a bore **120** in the elongated introducer sleeve **110**. By actuating the first and second handle portions, **114a** and **114b**, a working end **122** can be deployed from a first retracted position (FIG. 1) in the distal portion of bore **120** in introducer sleeve **110** to an extended position as shown in FIG. 2. In FIG. 2, it can be seen that the first and second handle portions, **114a** and **114b**, are in a second actuated position with the working end **122** deployed from the bore **120** in introducer sleeve **110**. (36) FIGS. 2 and 3 shows that ablation system **100** includes an RF energy source **130A** and RF controller **130B** in a control unit **135**. The RF energy source **130A** is connected to the hand-held device **105** by a flexible conduit **136** with a plug-in connector **137** configured with a gas inflow channel, a gas outflow channel, and first and second electrical leads for connecting to receiving connector **138** in the control unit **135**. The control unit **135**, as will be described further below in FIGS. 3 and 4, further comprises a neutral gas inflow source **140A**, gas flow controller **140B** and optional vacuum or negative pressure source **145** to provide controlled gas inflows and gas outflows to and from the working end **122**. The control unit **135** further includes a balloon inflation source **148** for inflating an expandable sealing balloon **225** carried on introducer sleeve **110** as described further below.

(37) Referring to FIG. 2, the working end **122** includes a flexible, thin-wall member or structure **150** of a dielectric material that when expanded has a triangular shape configured for contacting the patient's endometrial lining that is targeted for ablation. In one embodiment as shown in FIGS. 2, 5 and 6, the dielectric structure **150** comprises a thin-wall material such as silicone with a fluid-tight interior chamber **152**.

(38) In an embodiment, an expandable-collapsible frame assembly **155** is disposed in the interior chamber. Alternatively, the dielectric structure may be expanded by a neutral gas without a frame, but using a frame offers a number of advantages. First, the uterine cavity is flattened with the opposing walls in contact with one another. Expanding a balloon-type member may cause undesirable pain or spasms. For this reason, a flat structure that is expanded by a frame is better suited for deployment in the uterine cavity. Second, in embodiments herein, the neutral gas is converted to a conductive plasma at a very low pressure controlled by gas inflows and gas outflows—so that any pressurization of a balloon-type member with the neutral gas may exceed a desired pressure range and would require complex controls of gas inflows and gas outflows. Third, as described below, the frame provides an electrode for contact with the neutral gas in the interior chamber **152** of the dielectric structure **150**, and the frame **155** extends into all regions of the interior chamber to insure electrode exposure to all regions of the neutral gas and plasma. The frame **155** can be constructed of any flexible material with at least portions of the frame functioning as spring elements to move the thin-wall structure **150** from a collapsed configuration (FIG. 1) to an expanded, deployed configuration (FIG. 2) in a patient's uterine cavity. In one embodiment, the frame **155** comprises stainless steel elements **158a**, **158b** and **160a** and **160b** that function akin to leaf springs. The frame can be a stainless steel such as 316 SS, 17A SS, 420 SS, 440 SS or the frame can be a NiTi material. The frame preferably extends along a single plane, yet remains thin transverse to the plane, so that the frame may expand into the uterine cavity. The frame elements can have a thickness ranging from about 0.005" to 0.025". As can be seen in FIGS. 5 and 6, the proximal ends **162a** and **162b** of spring elements **158a**, **158b** are fixed (e.g., by welds **164**) to the distal end **165** of sleeve member **115**. The proximal ends **166a** and **166b** of spring elements **160a**, **160b** are welded to distal portion **168** of a secondary translatable sleeve **170** that can be extended from bore **175** in translatable sleeve **115**. The secondary translatable sleeve **170** is dimensioned for a loose fit in bore **175** to allow gas flows within bore **175**. FIGS. 5 and 6 further

illustrate the distal ends **176a** and **176b** of spring elements **158a**, **158b** are welded to distal ends **178a** and **178b** of spring elements **160a** and **160b** to thus provide a frame **155** that can be moved from a linear shape (see FIG. 1) to an expanded triangular shape (FIGS. 5 and 6).

(39) As will be described further below, the bore **175** in sleeve **115** and bore **180** in secondary translatable sleeve **170** function as gas outflow and gas inflow lumens, respectively. It should be appreciated that the gas inflow lumen can comprise any single lumen or plurality of lumens in either sleeve **115** or sleeve **170** or another sleeve, or other parts of the frame **155** or the at least one gas flow lumen can be formed into a wall of dielectric structure **150**. In FIGS. 5, 6 and 7 it can be seen that gas inflows are provided through bore **180** in sleeve **170**, and gas outflows are provided in bore **175** of sleeve **115**. However, the inflows and outflows can be also be reversed between bores **175** and **180** of the various sleeves. FIGS. 5 and 6 further show that a rounded bumper element **185** is provided at the distal end of sleeve **170** to ensure that no sharp edges of the distal end of sleeve **170** can contact the inside of the thin dielectric wall **150**. In one embodiment, the bumper element **185** is silicone, but it could also comprise a rounded metal element. FIGS. 5 and 6 also show that a plurality of gas inflow ports **188** can be provided along a length of in sleeve **170** in chamber **152**, as well as a port **190** in the distal end of sleeve **170** and bumper element **185**. The sectional view of FIG. 7 also shows the gas flow passageways within the interior of introducer sleeve **110**.

(40) It can be understood from FIGS. 1, 2, 5 and 6 that actuation of first and second handle portions, **114a** and **114b**, (i) initially causes movement of the assembly of sleeves **115** and **170** relative to bore **120** of introducer sleeve **110**, and (ii) secondarily causes extension of sleeve **170** from bore **175** in sleeve **115** to expand the frame **155** into the triangular shape of FIG. 5. The dimensions of the triangular shape are suited for a patient uterine cavity, and for example can have an axial length A ranging from 4 to 10 cm and a maximum width B at the distal end ranging from about 2 to 5 cm. In one embodiment, the thickness C of the thin-wall structure **150** can be from 1 to 4 mm as determined by the dimensions of spring elements **158a**, **158b**, **160a** and **160b** of frame assembly **155**. It should be appreciated that the frame assembly **155** can comprise round wire elements, flat spring elements, of any suitable metal or polymer that can provide opening forces to move thin-wall structure **150** from a collapsed configuration to an expanded configuration within the patient uterus. Alternatively, some elements of the frame **155** can be spring elements and some elements can be flexible without inherent spring characteristics.

(41) As will be described below, the working end embodiment of FIGS. 2, 5 and 6 has a thin-wall structure **150** that is formed of a dielectric material such as silicone that permits capacitive coupling of current to engaged tissue while the frame assembly **155** provides structural support to position the thin-wall structure **150** against tissue. Further, gas inflows into the interior chamber **152** of the thin-wall structure can assist in supporting the dielectric wall so as to contact endometrial tissue. The dielectric thin-wall structure **150** can be free from fixation to the frame assembly **155**, or can be bonded to an outward-facing portion or portions of frame elements **158a** and **158b**. The proximal end **182** of thin-wall structure **150** is bonded to the exterior of the distal end of sleeve **115** to thus provide a sealed, fluid-tight interior chamber **152** (FIG. 5).

(42) In one embodiment, the gas inflow source **140A** comprises one or more compressed gas cartridges that communicate with flexible conduit **136** through plug-in connector **137** and receiving connector **138** in the control unit **135** (FIGS. 1-2). As can be seen in FIGS. 5-6, the gas inflows from source **140A** flow through bore **180** in sleeve **170** to open terminations **188** and **190** therein to flow into interior chamber **152**. A vacuum source **145** is connected through conduit **136** and connector **137** to allow circulation of gas flow through the interior chamber **152** of the thin-wall dielectric structure **150**. In FIGS. 5 and 6, it can be seen that gas outflows communicate with vacuum source **145** through open end **200** of bore **175** in sleeve **115**. Referring to FIG. 5, it can be seen that frame elements **158a** and **158b** are configured with a plurality of apertures **202** to allow for gas flows through all interior portions of the frame elements, and thus gas inflows from open terminations **188**, **190** in bore **180** are free to circulated through interior chamber **152** to return to

an outflow path through open end **200** of bore **175** of sleeve **115**. As will be described below (see FIGS. **3-4**), the gas inflow source **140A** is connected to a gas flow or circulation controller **140B** which controls a pressure regulator **205** and also controls vacuum source **145** which is adapted for assisting in circulation of the gas. It should be appreciated that the frame elements can be configured with apertures, notched edges or any other configurations that allow for effective circulation of a gas through interior chamber **152** of the thin-wall structure **150** between the inflow and outflow passageways.

(43) Now turning to the electrosurgical aspects of the invention, FIGS. **5** and **6** illustrate opposing polarity electrodes of the system **100** that are configured to convert a flow of neutral gas in chamber **152** into a plasma **208** (FIG. **6**) and to allow capacitive coupling of current through a wall **210** of the thin-wall dielectric structure **150** to endometrial tissue in contact with the wall **210**. The electrosurgical methods of capacitively coupling RF current across a plasma **208** and dielectric wall **210** are described in U.S. patent application Ser. No. 12/541,043; filed Aug. 13, 2009 and U.S. application Ser. No. 12/541,050, referenced above. In FIGS. **5** and **6**, the first polarity electrode **215** is within interior chamber **152** to contact the neutral gas flow and comprises the frame assembly **155** that is fabricated of an electrically conductive stainless steel. In another embodiment, the first polarity electrode can be any element disposed within the interior chamber **152**, or extendable into interior chamber **152**. The first polarity electrode **215** is electrically coupled to sleeves **115** and **170** which extends through the introducer sleeve **110** to handle **106** and conduit **136** and is connected to a first pole of the RF source energy source **130A** and controller **130B**. A second polarity electrode **220** is external of the internal chamber **152** and in one embodiment the electrode is spaced apart from wall **210** of the thin-wall dielectric structure **150**. In one embodiment as depicted in FIGS. **5** and **6**, the second polarity electrode **220** comprises a surface element of an expandable balloon member **225** carried by introducer sleeve **110**. The second polarity electrode **220** is coupled by a lead (not shown) that extends through the introducer sleeve **110** and conduit **136** to a second pole of the RF source **130A**. It should be appreciated that second polarity electrode **220** can be positioned on sleeve **110** or can be attached to surface portions of the expandable thin-wall dielectric structure **150**, as will be described below, to provide suitable contact with body tissue to allow the electrosurgical ablation of the method of the invention. The second polarity electrode **220** can comprise a thin conductive metallic film, thin metal wires, a conductive flexible polymer or a polymeric positive temperature coefficient material. In one embodiment depicted in FIGS. **5** and **6**, the expandable member **225** comprises a thin-wall compliant balloon having a length of about 1 cm to 6 cm that can be expanded to seal the cervical canal. The balloon **225** can be inflated with a gas or liquid by any inflation source **148**, and can comprise a syringe mechanism controlled manually or by control unit **135**. The balloon inflation source **148** is in fluid communication with an inflation lumen **228** in introducer sleeve **110** that extends to an inflation chamber of balloon **225** (see FIG. **7**).

(44) Referring back to FIG. **1**, the control unit **135** can include a display **230** and touch screen or other controls **232** for setting and controlling operational parameters such as treatment time intervals, treatment algorithms, gas flows, power levels and the like. Suitable gases for use in the system include argon, other noble gases and mixtures thereof. In one embodiment, a footswitch **235** is coupled to the control unit **135** for actuating the system.

(45) The box diagrams of FIGS. **3** and **4** schematically depict the system **100**, subsystems and components that are configured for an endometrial ablation system. In the box diagram of FIG. **3**, it can be seen that RF energy source **130A** and circuitry is controlled by a controller **130B**. The system can include feedback control systems that include signals relating to operating parameters of the plasma in interior chamber **152** of the dielectric structure **150**. For example, feedback signals can be provided from at least one temperature sensor **240** in the interior chamber **152** of the dielectric structure **150**, from a pressure sensor within, or in communication, with interior chamber **152**, and/or from a gas flow rate sensor in an inflow or outflow channel of the system. FIG. **4** is a

schematic block diagram of the flow control components relating to the flow of gas media through the system **100** and hand-held device **105**. It can be seen that a pressurized gas source **140A** is linked to a downstream pressure regulator **205**, an inflow proportional valve **246**, flow meter **248** and normally closed solenoid valve **250**. The valve **250** is actuated by the system operator which then allows a flow of a neutral gas from gas source **140A** to circulate through flexible conduit **136** and the device **105**. The gas outflow side of the system includes a normally open solenoid valve **260**, outflow proportional valve **262** and flow meter **264** that communicate with vacuum pump or source **145**. The gas can be exhausted into the environment or into a containment system. A temperature sensor **270** (e.g., thermocouple) is shown in FIG. **4** that is configured for monitoring the temperature of outflow gases. FIG. **4** further depicts an optional subsystem **275** which comprises a vacuum source **280** and solenoid valve **285** coupled to the controller **140B** for suctioning steam from a uterine cavity **302** at an exterior of the dielectric structure **150** during a treatment interval. As can be understood from FIG. **4**, the flow passageway from the uterine cavity **302** can be through bore **120** in sleeve **110** (see FIGS. **2**, **6** and **7**) or another lumen in a wall of sleeve **110** can be provided.

(46) FIGS. **8A-8D** schematically illustrate a method of the invention wherein (i) the thin-wall dielectric structure **150** is deployed within a patient uterus and (ii) RF current is applied to a contained neutral gas volume in the interior chamber **152** to contemporaneously create a plasma **208** in the chamber and capacitively couple current through the thin dielectric wall **210** to apply ablative energy to the endometrial lining to accomplish global endometrial ablation.

(47) More in particular, FIG. **8A** illustrates a patient uterus **300** with uterine cavity **302** surrounded by endometrium **306** and myometrium **310**. The external cervical os **312** is the opening of the cervix **314** into the vagina **316**. The internal os or opening **320** is a region of the cervical canal that opens to the uterine cavity **302**. FIG. **8A** depicts a first step of a method of the invention wherein the physician has introduced a distal portion of sleeve **110** into the uterine cavity **302**. The physician gently can advance the sleeve **110** until its distal tip contacts the fundus **324** of the uterus. Prior to insertion of the device, the physician can optionally introduce a sounding instrument into the uterine cavity to determine uterine dimensions, for example from the internal os **320** to fundus **324**.

(48) FIG. **8B** illustrates a subsequent step of a method of the invention wherein the physician begins to actuate the first and second handle portions, **114a** and **114b**, and the introducer sleeve **110** retracts in the proximal direction to expose the collapsed frame **155** and thin-wall structure **150** within the uterine cavity **302**. The sleeve **110** can be retracted to expose a selected axial length of thin-wall dielectric structure **150**, which can be determined by markings **330** on sleeve **115** (see FIG. **1**) which indicate the axial travel of sleeve **115** relative to sleeve **170** and thus directly related to the length of deployed thin-wall structure **150**. FIG. **2** depicts the handle portions **114a** and **114b** fully approximated thus deploying the thin-wall structure to its maximum length.

(49) FIG. **8C** illustrates several subsequent steps of a method of the invention. FIG. **8C** first depicts the physician continuing to actuate the first and second handle portions, **114a** and **114b**, which further actuates the frame **155** (see FIGS. **5-6**) to expand the frame **155** and thin-wall structure **150** to a deployed triangular shape to contact the patient's endometrial lining **306**. The physician can slightly rotate and move the expanding dielectric structure **150** back and forth as the structure is opened to insure it is opened to the desired extent. In performing this step, the physician can actuate handle portions, **114a** and **114b**, a selected degree which causes a select length of travel of sleeve **170** relative to sleeve **115** which in turn opens the frame **155** to a selected degree. The selected actuation of sleeve **170** relative to sleeve **115** also controls the length of dielectric structure deployed from sleeve **110** into the uterine cavity. Thus, the thin-wall structure **150** can be deployed in the uterine cavity with a selected length, and the spring force of the elements of frame **155** will open the structure **150** to a selected triangular shape to contact or engage the endometrium **306**. In one embodiment, the expandable thin-wall structure **150** is urged toward and maintained in an open

position by the spring force of elements of the frame **155**. In the embodiment depicted in FIGS. **1** and **2**, the handle **106** includes a locking mechanism with finger-actuated sliders **332** on either side of the handle that engage a grip-lock element against a notch in housing **333** coupled to introducer sleeve **110** (FIG. **2**) to lock sleeves **115** and **170** relative to introducer sleeve **110** to maintain the thin-wall dielectric structure **150** in the selected open position.

(50) FIG. **8C** further illustrates the physician expanding the expandable balloon structure **225** from inflation source **148** to thus provide an elongated sealing member to seal the cervix **314** outward from the internal os **320**. Following deployment of the thin-wall structure **150** and balloon **225** in the cervix **314**, the system **100** is ready for the application of RF energy to ablate endometrial tissue **306**. FIG. **8C** next depicts the actuation of the system **100**, for example, by actuating footswitch **235**, which commences a flow of neutral gas from source **140A** into the interior chamber **152** of the thin-wall dielectric structure **150**. Contemporaneous with, or after a selected delay, the system's actuation delivers RF energy to the electrode arrangement which includes first polarity electrode **215** (+) of frame **155** and the second polarity electrode **220** (−) which is carried on the surface of expandable balloon member **225**. The delivery of RF energy delivery will instantly convert the neutral gas in interior chamber **152** into conductive plasma **208** which in turn results in capacitive coupling of current through the dielectric wall **210** of the thin-wall structure **150** resulting in ohmic heating of the engaged tissue. FIG. **8C** schematically illustrates the multiplicity of RF current paths **350** between the plasma **208** and the second polarity electrode **220** through the dielectric wall **210**. By this method, it has been found that ablation depths of three mm to six mm or more can be accomplished very rapidly, for example in 60 seconds to 120 seconds dependent upon the selected voltage and other operating parameters. In operation, the voltage at which the neutral gas inflow, such as argon, becomes conductive (i.e., converted in part into a plasma) is dependent upon a number of factors controlled by the controllers **130B** and **140B**, including the pressure of the neutral gas, the volume of interior chamber **152**, the flow rate of the gas through the chamber **152**, the distance between electrode **210** and interior surfaces of the dielectric wall **210**, the dielectric constant of the dielectric wall **210** and the selected voltage applied by the RF source **130**, all of which can be optimized by experimentation. In one embodiment, the gas flow rate can be in the range of 5 ml/sec to 50 ml/sec. The dielectric wall **210** can comprise a silicone material having a thickness ranging from a 0.005" to 0.015 and having a relative permittivity in the range of 3 to 4. The gas can be argon supplied in a pressurized cartridge which is commercially available. Pressure in the interior chamber **152** of dielectric structure **150** can be maintained between 14 psia and 15 psia with zero or negative differential pressure between gas inflow source **140A** and negative pressure or vacuum source **145**. The controller is configured to maintain the pressure in interior chamber in a range that varies by less than 10% or less than 5% from a target pressure. The RF power source **130A** can have a frequency of 450 to 550 KHz, and electrical power can be provided within the range of 600 Vrms to about 1200 Vrms and about 0.2 Amps to 0.4 Amps and an effective power of 40 W to 100 W. In one method, the control unit **135** can be programmed to delivery RF energy for a preselected time interval, for example, between 60 seconds and 120 seconds. One aspect of a treatment method corresponding to the invention consists of ablating endometrial tissue with RF energy to elevate endometrial tissue to a temperature greater than 45 degrees Celsius for a time interval sufficient to ablate tissue to a depth of at least 1 mm. Another aspect of the method of endometrial ablation of consists of applying radiofrequency energy to elevate endometrial tissue to a temperature greater than 45 degrees Celsius without damaging the myometrium.

(51) FIG. **8D** illustrates a final step of the method wherein the physician deflates the expandable balloon member **225** and then extends sleeve **110** distally by actuating the handles **114a** and **114b** to collapse frame **155** and then retracting the assembly from the uterine cavity **302**. Alternatively, the deployed working end **122** as shown in FIG. **8C** can be withdrawn in the proximal direction from the uterine cavity wherein the frame **155** and thin-wall structure **150** will collapse as it is pulled through the cervix. FIG. **8D** shows the completed ablation with the ablated endometrial

tissue indicated at **360**.

(52) In another embodiment, the system can include an electrode arrangement in the handle **106** or within the gas inflow channel to pre-ionize the neutral gas flow before it reaches the interior chamber **152**. For example, the gas inflow channel can be configured with axially or radially spaced apart opposing polarity electrodes configured to ionize the gas inflow. Such electrodes would be connected in separate circuitry to an RF source. The first and second electrodes **215** (+) and **220** (−) described above would operate as described above to provide the current that is capacitively coupled to tissue through the walls of the dielectric structure **150**. In all other respects, the system and method would function as described above.

(53) Now turning to FIGS. **9** and **10**, an alternate working end **122** with thin-wall dielectric structure **150** is shown. In this embodiment, the thin-wall dielectric structure **150** is similar to that of FIGS. **5** and **6** except that the second polarity electrode **220'** that is exterior of the internal chamber **152** is disposed on a surface portion **370** of the thin-wall dielectric structure **150**. In this embodiment, the second polarity electrode **220'** comprises a thin-film conductive material, such as gold, that is bonded to the exterior of thin-wall material **210** along two lateral sides **354** of dielectric structure **150**. It should be appreciated that the second polarity electrode can comprise one or more conductive elements disposed on the exterior of wall material **210**, and can extend axially, or transversely to axis **111** and can be singular or multiple elements. In one embodiment shown in more detail in FIG. **10**, the second polarity electrode **220'** can be fixed on another lubricious layer **360**, such as a polyimide film, for example KAPTON®. The polyimide tape extends about the lateral sides **354** of the dielectric structure **150** and provides protection to the wall **210** when it is advanced from or withdrawn into bore **120** in sleeve **110**. In operation, the RF delivery method using the embodiment of FIGS. **9** and **10** is the same as described above, with RF current being capacitively coupled from the plasma **208** through the wall **210** and endometrial tissue to the second polarity electrode **220'** to cause the ablation.

(54) FIG. **9** further shows an optional temperature sensor **390**, such as a thermocouple, carried at an exterior of the dielectric structure **150**. In one method of use, the control unit **135** can acquire temperature feedback signals from at least one temperature sensor **390** to modulate or terminate RF energy delivery, or to modulate gas flows within the system. In a related method of the invention, the control unit **135** can acquire temperature feedback signals from temperature sensor **240** in interior chamber **152** (FIG. **6** to modulate or terminate RF energy delivery or to modulate gas flows within the system.

(55) In another embodiment of the invention, FIGS. **11-14** depict systems and methods for evaluating the integrity of the uterine cavity which may be perforated or otherwise damaged by the transcervical introduction of probes and instruments into a uterine cavity. If the uterine wall is perforated, it would be preferable to defer any ablation treatment until the uterine wall is healed. A method of the invention comprises introducing transcervically a probe into a patient's uterine cavity, providing a flow of a fluid (e.g., CO₂) through the probe into the uterine cavity and monitoring the rate of the flow to characterize the uterine cavity as perforated or non-perforated based on a change in the flow rate. If the flow rate drops to zero or close to zero, this indicates that the uterine cavity is intact and not perforated. If the flow rate does not drop to zero or close to zero, this indicates that a fluid flow is leaking through a perforation in the uterine cavity **302** into the uterine cavity or escaping around an occlusion balloon that occludes the cervical canal.

(56) In FIG. **11**, it can be seen how a pressurized fluid source **405** and controller **410** for controlling and monitoring flows is in fluid communication with lumen **120** of introducer sleeve **110** (see FIG. **7**). In one embodiment, the fluid source can be a pressurized cartridge containing CO₂ or another biocompatible gas. In FIG. **12**, it can be seen that fluid source **405** communicates with a flexible conduit **412** that is connected to a “pig-tail” tubing connector **414** extending outward from handle **106** of the hand-held probe. A tubing in the interior of handle component **114a** provides a flow passageway **415** to the lumen **120** in the introducer sleeve. In another embodiment, the fluid source

405 and flexible conduit **408** can be integrated into conduit **136** of FIG. 1.

(57) In FIG. 11, it can be seen that the flow of fluid is introduced into the uterine cavity **302** after the balloon **225** in the cervical canal has been inflated and after the working end and dielectric structure **150** has been expanded into its triangular shape to occupy the uterine cavity. Thus, the CO₂ gas flows around the exterior surfaces of expanded dielectric structure **150** to fill the uterine cavity. Alternatively, the flow of CO₂ can be provided after the balloon **225** in the cervical canal is inflated but before the dielectric structure **150** is expanded.

(58) FIG. 13 is a block diagram that schematically depicts the components of subsystem **420** that provides the flow of CO₂ to and through the hand-held probe **105**. It can be seen that pressurized fluid source **405** communicates with a downstream pressure regulator **422**, a proportional valve **424**, flow meter **440**, normally closed solenoid valve **450** and one-way valve **452**. The valve **450** upon actuation by the system operator allows a flow of CO₂ gas from source **405** at a predetermined flow rate and pressure through the subsystem and into the uterine cavity **302**.

(59) In one embodiment of the method of operation, the physician actuates the system and electronically opens valve **450** which can provide a CO₂ flow through the system. The controller **410** monitors the flow meter or sensor **440** over an interval that can range from 1 second to 60 seconds, or 5 second to 30 seconds to determine the change in the rate of flow and/or a change in the rate of flow. In an embodiment, the flow sensor comprises a Honeywell AWM5000 Series Mass Airflow Sensor, for example Model AWM5101, that measure flows in units of mass flow. In one embodiment, the initial flow rate is between 0.05 slpm (standard liters per minute) and 2.0 slpm, or between 0.1 slpm and 0.2 slpm. The controller **410** includes a microprocessor or programmable logic device that provides a feedback signal from the flow sensors indicating either (i) that the flow rate has dropped to zero or close to zero to thus characterize the uterine cavity as non-perforated, or (ii) that the flow rate has not dropped to a predetermined threshold level within a predetermined time interval to thus characterize the uterine cavity as perforated or that there is a failure in occlusion balloon **225** or its deployment so that the cervical canal is not occluded. In one embodiment, the threshold level is 0.05 slpm for characterizing the uterine cavity as non-perforated. In this embodiment, the controller provides a signal indicating a non-perforated uterine cavity if the flow drops below 0.05 slpm between the fifth second of the flow and the flow time-out, which can be, for example, 30 seconds.

(60) FIG. 14 depicts aspects of an algorithm used by controller **410** to accomplish a uterine cavity integrity check, with the first step comprising actuating a footswitch or hand switch. Upon actuation, a timer is initialized for 1 to 5 seconds to determine that a fluid source **405** is capable of providing a fluid flow, which can be checked by a pressure sensor between the source **405** and pressure regulator **422**. If no flow is detected, an error signal is provided, such as a visual display signal on the control unit **135** (FIG. 1).

(61) As can be understood from FIG. 14, after the fluid source **405** is checked, the controller opens the supply solenoid valve **450** and a timer is initialized for a 1 to 5 second test interval to insure fluid flows through the subsystem **420** of FIG. 13, with either or both a flow meter **440** or a pressure sensor. At the same time as valve **450** is opened, a timer is initialized for cavity integrity test interval of 30 seconds. The controller **410** monitors the flow meter **440** and provides a signal characterizing the uterine cavity as non-perforated if, at any time after the initial 5 second check interval and before the end of the timed-out period (e.g., the 30 second time-out), the flow rate drops below a threshold minimum rate, in one embodiment, to below 0.05 slpm. If the interval times out after 30 seconds and the flow rate does not drop below this threshold, then a signal is generated that characterizes that the uterine cavity is perforated. This signal also can indicate a failure of the occlusion balloon **225**.

(62) Referring to FIG. 14, in one embodiment, in response or otherwise as a result of the signal that the uterine cavity is not perforated, the controller **410** can automatically enable and activate the RF ablation system described above to perform an ablation procedure. The controller **410** can provide

a time interval from 1 to 15 seconds to allow CO₂ gas to vent from the uterine cavity **302** before activating RF energy delivery. In another embodiment, the endometrial ablation system may include the optional subsystem **275** for exhausting fluids or gas from the uterine cavity during an ablation treatment (see FIG. **4** and accompanying text). This subsystem **275** can be actuated to exhaust CO₂ from the uterine cavity **302** which include opening solenoid valve **285** shown in FIG. **4**.

(63) The system can further include an override to repeat the cavity integrity check, for example, after evaluation and re-deployment of the occlusion balloon **225**.

(64) FIGS. **15** and **16** represent another system and method for characterizing the uterine cavity as being non-perforated so as to safely permit an ablation procedure. This system and method utilizes variations in the algorithms that introduce a gas media fluid into the uterine cavity and thereafter measure the changes in flow rates in the gas media. The system again is configured to introduce a gas into the uterine cavity after deployment and expansion of an ablation device in the cavity. If the flow rate drops of the gas to approximately zero, this indicates that the uterine cavity is intact and not perforated. In the event, the flow rate of the gas does not drop, there is likely a gas flow escaping from the uterine cavity **302** through a perforation in the uterine wall.

(65) FIG. **15** schematically illustrates three different conditions that may occur when operating the system, which indicate whether the system is functioning properly, and whether the uterine wall is non-perforated or perforated. In FIG. **15**, the vertical axis indicates a gas flow rate measure in slpm (standard liters per minute), and the horizontal axis represents time in seconds. In one system variation, a gas source **405** such as a pressurized cartridge containing CO₂ is controlled by a controller **410**, and the gas is introduced into the uterine cavity through a passageway in the device introducer sleeve **110** as described above (FIGS. **11-13**). The controller **410** and flowmeter monitors flows from the device into the uterine cavity (FIG. **13**). The initial flow rate can be in the range of 0.010 slpm to 0.20 slpm. In one aspect of the invention, a minimum flow rate has been found to be important as a system diagnostic check to insure gas flow is reaching the uterine cavity. Thus, FIG. **15** illustrates gas flow rate curve in a “condition 1” that may occur when the system fails in delivering gas through the passageways of the system. In one variation, the “condition 1” will be represented by a flow rate over time wherein the flow rate does not achieve a minimum threshold flow rate, which can be from 0.010 slpm to 0.050 slpm over a predetermined time interval. In one variation, the minimum flow rate is 0.035 slpm. The time interval can be from 1 second to 15 seconds. This “condition 1” as in FIG. **15** could occur, for example, if the gas supply tubing within the device were kinked or pinched which would then prevent gas flow through the system and into the uterine cavity. In a related variation that indicates system failure, a controller algorithm can calculate the volume of gas delivered, and if the volume is less than a threshold volume, then a system failure or fault can be determined. The gas volume V_{sub.1} is represented by the “area under the curve” in FIG. **15**, which is a function of flow rate and time.

(66) FIG. **15** further illustrates a flow rate curve in a “condition 2” which corresponds to an intact, non-perforated uterine cavity. As can be understood from a practical perspective, a gas flow into an intact uterine cavity at a set pressure from a low pressure source, for example within a range of 0.025 psi to 1.0 psi, would provide an increasing flow rate into the cavity until the cavity was filled with gas, and thereafter the flow rate would diminish to a very low or zero flow rate. Such a “condition 2” flow rate curve as in FIG. **15** further assumes that there is an adequate sealing mechanism in the cervical canal. Thus, if controller obtains flow rate data from the flowmeter indicating “condition 2”, then the patient's uterus is non-perforated and is suitable for an ablation. In operation, the controller can look at various specific aspects and parameters of the flow rate curve of “condition 2” in FIG. **15** to determine that the uterine cavity integrity test has passed, wherein such parameters can comprise any single parameter or a combination of the following parameters: (i) the flow rate falling below a threshold rate, for example between 0.010-0.10 slpm; (ii) a change in rate of flow; (iii) a peak flow rate; (iii) the total gas volume V_{sub.2} delivered; (iv)

an actual flow rate at a point in time compared to a peak flow rate; (v) a derivative of flow rate at a point in time, and (vi) any of the preceding parameters combined with a predetermined time interval. In one embodiment, a constant pressure (0.85 psi) gas is introduced and a minimum threshold flow is set at 0.035 slpm. A peak flow is calculated after a time interval of 2 to 15 seconds, and thereafter it is determined if the flow rate diminished by at least 10%, 20%, 30%, 40% or 50% over a time interval of less than 30 seconds.

(67) FIG. 15 next illustrates a flow rate curve in “condition 3” which represents a gas flow when there is a perforated wall in a uterine cavity, which would allow the gas to escape into the abdominal cavity. In FIG. 15, a gas flow at a constant pressure is shown ramping up in flow rate until it levels off and may decline but not the rate of decline to may not go below a threshold value or may not decline a significant amount relative to a peak flow rate. Such a flow rate curve over time would indicate that the gas is leaking from the uterine cavity.

(68) Now turning to FIG. 16, an algorithm diagram is shown that describe one variation in a method of operating a uterine cavity integrity test based on measuring gas flow rates over a selected time interval. At the top of the diagram, the physician actuates the system in which a valve 450 is opened to provide a CO.sub.2 flow through the system (FIG. 14). The controller 410 provides a flow at a pressure, for example 0.85 psi. The actuation of the system also starts a timer wherein a first interval is 30 seconds or less. Over this 30-second interval, the controller records the peak flow rate which typically can occur within 2 to 10 seconds, then monitors the flow rate over the remainder of the 30 second interval and determined whether the flow rate drops 20% or more from the peak flow rate. Then, the controller additionally monitors whether the flow rate falls below a threshold value, for example 0.035 slpm. If these two conditions are met, the test indicates that there is no leakage of gas media from the uterine cavity. If the flow rates does not drop 20% from its peak with 30 seconds together with flow being below threshold value, then the test fails indicating a leak of gas from the uterine cavity. Thereafter, the diagram in FIG. 16, indicates one additional test which consists of calculating the volume of gas delivered and comparing the volume to the maximum volume within a kinked gas delivery line. If the delivered gas volume is less than the capacity of the gas delivery line, then the test fails and the signal on the controller can indicate this type of test failure. If the delivered gas volume is greater than the capacity of a gas delivery line, then the test passes. In one variation of the controller algorithm can then automatically actuate the delivery of RF energy in an ablation cycle. Alternatively, the controller can provide a signal that the test has passed, and the physician can manually actuate the RF ablation system.

(69) FIG. 17 schematically illustrates another system and method for characterizing integrity of the walls of a uterine cavity. As can be seen in FIG. 17, an introducer sleeve 510 carrying an expandable working end 520 in deployed in the uterine cavity 302. The working end includes a balloon-like member 522 with a fluid-tight interior chamber 524. In one embodiment, the working end 510 is expanded laterally by frame elements 526a and 526b, which is similar to previously described embodiments. In addition, a pressurized gas source 540 is actuated to provide an inflation gas thru interior sleeve 542 and ports 544 therein that further expands and opens the working end 520 transverse to opening forces applied by frame elements 526a and 526b. The inflation gas can comprise an argon gas that later is converted to a plasma as described previously. The inflation gas can pressurize the working end to a selected pressure ranging from 0.10 psi to 10 psi. In one variation, the pressure can be 0.50 psi.

(70) As can be seen in FIG. 17, an expandable member 548 or balloon is expanded to prevent any gas flow outwardly through the bore 550 in introducer sleeve 510. Thereafter, a gas inflow system 410 similar to that of FIG. 13 is utilized to flow a gas source, such as CO.sub.2 into the uterine cavity 302 (FIG. 17). In FIG. 17, the gas inflow is indicated by arrows 555 which can comprise an inflow at a predetermined pressure through passageway 558 as described above, and in one variation can be 0.85 psi. The test for uterine cavity integrity then can monitor one or more gas leakage parameters relating to the inflation gas in the interior chamber 524 of the working end 520.

For example, the flow into the uterine cavity **302** will cause an outflow of gas from the interior chamber **524** through passageway **558** which can be measure by a flow meter, or the volume of gas outflow can be measured or the change in gas pressure can be measured. If there is no leak in the uterine cavity, the parameter of the inflation can in the interior chamber **524** will reach an equilibrium in relation to the CO.sub.2 inflow into the cavity. If the inflation gas parameter does not reach an equilibrium, then the change in parameter (flow, volume or pressure) will indicate a leakage of gas from the uterine cavity through a perforation. In general, a method of characterizing the integrity of a patient's uterus comprises positioning a probe working end in a patient's uterine cavity, the working end comprising an inflated resilient structure, introducing a flow of a gas through the probe into a uterine cavity exterior about the exterior of the working end, and measuring a gas flow, gas volume or gas pressure parameter of the inflation media in the inflated resilient structure in response to the gas flow into the uterine cavity.

(71) FIGS. **18A-20B** illustrates other methods of characterizing and/or treating a patient's uterus, which include a multi-stage test for uterine wall integrity which enhances safety. In general, a multi-stage test corresponding to the invention comprises positioning a probe working end in a patient's uterine cavity, introducing a first fluid into the expandable working end, introducing a second fluid into the uterine cavity exterior of the working end, and performing first and second monitoring tests relating to parameters of the first and second fluids to thereby characterize a uterine wall. Thereafter, the physician can actuate an ablation mechanism carried by the working end upon characterization of the uterine wall as intact or non-perforated. In one embodiment, the step of actuating the ablation mechanism is automated by a controller upon a signal from at least one sensor that uterine cavity is intact.

(72) FIGS. **18A** and **18B** illustrate in more detail a two-stage method for testing the integrity of a uterine wall or uterine cavity. In FIG. **18A**, it can be seen that the elongate introducer sleeve **510** carrying an expandable working end **520** is properly deployed in the uterine cavity **302**. The working end again includes an expandable thin-wall resilient member such as a dielectric member **522** with a fluid-tight interior chamber **524**. The cervical seal **604** is positioned in the cervical canal with a cervical cuff **606** expanded and the dielectric structure **522** is expanded as described previously. FIG. **18A** further illustrates an inflow of CO.sub.2 gas from source **420** into the uterine cavity **302** through sleeve **510** about the exterior of the dielectric structure **522**. Thus, the first test stage illustrated in FIG. **18A** consists of the cavity integrity test described above in conjunction with FIGS. **15-16** wherein the CO.sub.2 inflow is monitored for a predetermined decay in the CO.sub.2 flow rate to determine whether a perforation in the uterine cavity may exist.

(73) Now turning to FIG. **18B**, a second stage of the test is illustrated. In the second stage, Argon gas from source **610** is controlled by the controller **615** to provide a flow into the interior chamber **524** of the dielectric structure **522**. As can be understood in FIG. **18B**, any significant or slight expansion of the dielectric surface **612** (see arrows in FIG. **18B**) by the Argon inflation in the dielectric structure **522** will impinge on CO.sub.2 flows about the exterior of the dielectric structure **522**. Thus, the second stage of the cavity integrity test includes monitoring the flow rate of the CO.sub.2 with a flowmeter **625** for changes that are indicative of expansion of the dielectric structure wall which impinges on the CO.sub.2 flow rate. In FIG. **18B**, it can be understood that when the dielectric structure **522** is fully expanded within the uterine cavity, there remains very little space between the dielectric wall and the uterine wall for CO.sub.2 to flow. In other words, any expansion of the dielectric structure **522** will result in a significant change in flow of the CO.sub.2. In some cases, the flow meter coupled to the CO.sub.2 inflow lumen can detect CO.sub.2 outflow. In the event the system is configured for a circulating flow of CO.sub.2 into and out of the dielectric structure, a flowmeter **625** can detect a change in the flow rate caused by the dielectric structure impinging on such a flow rate. The second stage of the test illustrated FIG. **18B** this can confirm dielectric structure **522** is properly deployed and expanded cavity **302**.

(74) FIG. **19** is a box diagram that indicates one [embodiment] mentioned and the steps involved in

the second stage of the two-stage cavity integrity test. In one embodiment, CO.sub.2 flow to the exterior of the dielectric structure is activated at T=zero. At T=2 seconds, the controller **615** activates the Argon positive pressure source at a flow rate of 0.8 SLPM using a flow control loop. Pressure for the Argon in the interior chamber the dielectric structure can be set at 0.5 psig using the proportional valve in the return line and a pressure sensing mechanism. Thereafter, the controller monitors CO.sub.2 flows to determine whether the Argon flow into the interior dielectric structure impinges on CO.sub.2 flows. In one example, after 2 seconds, a significant change the CO.sub.2 flows will indicate that the dielectric structure is substantially positioned in the uterine cavity **302** and is properly expanded. In this stage of the test, if the controller **615** does not receive a signal indicating a change in CO.sub.2 flows, this indicates that the dielectric structure **522** it is not deployed substantially within the uterine cavity **302** and the working end may have penetrated the wall of the uterine cavity to some extent and further is plugging the perforation thus preventing any leak.

(75) FIGS. **20A** and **20B** illustrate the two-stage cavity integrity test in a situation wherein the dielectric structure **522** has at least partially perforated the uterine wall, unlike the proper working end deployment as depicted in FIGS. **18A-18B**. As can be seen in FIG. **20A**, the dielectric structure **522** has penetrated the fundus **618** and is only partially opened. Such a uterine wall perforation could occur in the fundus **618** or elsewhere in the uterine wall in an asymmetrically shaped uterine cavity. Such a perforation could be caused by the working end of the probe itself, or the perforation could be caused by a sounding instrument that is used in a preliminary step in which the physician measures the length of the uterine cavity.

(76) FIG. **20A** illustrates the first stage of the two-part test wherein CO.sub.2 flows from the CO.sub.2 source **620** into the uterine cavity **302** through the introducer sleeve **510** and about the exterior of the dielectric structure **522**. It can be seen that the gas flows may not penetrate the perforation since the perforation may be effectively plugged by the silicone surface of the dielectric structure **522**. Thus, FIG. **20A** illustrates a condition wherein the first stage of the test which monitors only CO.sub.2 flow would indicate that the cavity has no perforations, when in fact there is a perforation that is masked by the dielectric structure **522** plugging the perforation.

(77) FIG. **20B** illustrates the second stage of the two-stage test wherein Argon is introduced into the interior chamber **524** the dielectric structure **522**. In FIG. **20B**, it can be seen that the expansion of the dielectric wall is limited as indicated by the arrows since the distal portion **632** portion of the dielectric structure **522** is embedded and not expandable within the perforation in the uterine fundus. In this case, the introduction of Argon gas into the dielectric structure detects the perforation in the uterine wall which otherwise would not have been detected by the first stage of the test. In the situation indicated in FIG. **20B**, the flow of Argon would be constrained and there would be little fluctuation in the CO.sub.2 flow rate—thus indicating that the dielectric structure **522** is not properly expanded and likely is disposed within a perforated uterine wall.

(78) FIG. **21** illustrates another variation in the second stage of the two-stage cavity integrity test that records and compares alternative parameters of fluid flows within the uterine cavity **302** and in the interior chamber **524** of dielectric structure **522**. This method variation compares a change in a gas pressure parameter in Argon gas in the dielectric structure **522** over a time interval of several seconds. First, the controller **615** turns on the Argon gas for 2 seconds at a flow rate of 0.8 SLPM using a flow control loop. The controller further sets the dielectric pressure at 0.5 psig. Next, the controller turns on the CO.sub.2 flow to the uterine cavity exterior of the dielectric structure. The controller **615** is configured to insure that the CO.sub.2 flow is sufficiently low to maintain a seal between the cervical cuff **606** and the interior os of the cervical canal. If CO.sub.2 flow were higher than a predetermined limit, the uterine integrity test could fail because of CO.sub.2 leakage around the cervical cuff.

(79) Next, the controller **615** turns off the Argon flow by closing a valve in the Argon flow system. Thus, the dielectric is maintained at 0.5 psig. After 2 seconds, the Argon pressure (P.sub.1) is

recorded with CO.sub.2 flowing about the exterior of the dielectric. Next, the CO.sub.2 flow is turned off and the Argon pressure is recorded after one second, which is Argon pressure P.sub.2 with no CO.sub.2 flowing about the exterior of the dielectric. The final step then compares P.sub.1 and P.sub.2. If P.sub.1 is greater than P.sub.2 (plus a predetermined margin) the cavity integrity test is successful and characterizes the uterine wall as non-perforated. If P.sub.1 is not greater than P.sub.2 (plus the predetermined margin), the cavity integrity test is not successful and a perforation detected message is displayed by the controller.

(80) In general, a method corresponding to the invention for characterizing a patient's uterus, comprises positioning an expandable structure in a patient's uterine cavity, introducing a gas into the uterine cavity exterior of expandable structure, introducing a gas into the expandable structure, and monitoring a gas parameter in the gas both interior and exterior of the expandable structure to thereby characterize the uterine cavity as either perforated or non-perforated. The gas introduced at the exterior of the expandable structure can be CO.sub.2. The gas introduced into the interior of the expandable structure can be a neutral gas. The method can monitor any gas parameter which is useful for leak detection. In one variation, the leak detecting parameter is a gas flow rate. In another variation, the leak detecting parameter is a gas pressure. In another variation, the leak detecting parameter is gas volume.

(81) In general, the method can include monitoring the gas parameters contemporaneously and/or sequentially. The monitoring step can monitor gas parameters first in the uterine cavity and then subsequently in the expandable structure, or vice versa. Further, the method can monitor a gas parameter in the uterine cavity at least twice and can monitor a gas parameter in the expandable structure at least twice.

(82) Now turning to FIGS. 22A-22B, another variation of system and method for operating a uterine cavity integrity test is based on measuring a gas inflow rate into the uterine cavity. FIG. 22A depicts an expandable working end or structure **700** that is operatively coupled to subsystems described previously, i.e., a pressurized CO.sub.2 flow source **705** for providing gas inflow into the uterine cavity **302**, a controller **710** for controlling gas inflows, and a flowmeter **715** operatively coupled to the controller for measuring the rate of gas inflows. In this embodiment, an additional negative pressure source **720** is provided for applying negative pressure to an interior chamber **722** of the thin-wall dielectric sheath **725** that again comprises a working end **700** similar to embodiments described above. The working end of FIGS. 22A-22B again includes an expandable frame **726** as in the FIGS. 5-6 that is expandable within interior chamber **722** of thin-wall dielectric sheath **725**. In FIG. 22A, the introducer sleeve **510** carrying the expandable working end **700** is properly deployed in uterine cavity **302**. The cervical seal **604** is positioned in the cervical canal with a cervical cuff **606** expanded and the dielectric sheath **725** is expanded as described previously. FIG. 22A further illustrates an inflow of CO.sub.2 gas from source **705** into the uterine cavity **302** through sleeve **510** about the exterior of the dielectric structure or sheath **725**.

(83) The test depicted in FIG. 22A is similar to the cavity integrity test described above in conjunction with FIGS. 15-16 wherein the CO.sub.2 inflow is monitored for a predetermined decay in the CO.sub.2 flow rate to determine whether a perforation in the uterine cavity may exist. The variation of the test in FIG. 22A adds an additional intermediate step. Prior to initiating a CO.sub.2 inflow, the controller **715** and test algorithm is configured to actuate negative pressure source **720** to thereby suction gas from the interior chamber **722** of dielectric sheath **725** to suction the thin-wall sheath against the interior frame **726**. The negative pressure can be from 5 to 10 psi below ambient, or any similar achievable negative pressure which is maintained for the subsequent step of inflowing CO.sub.2 into the uterine cavity. FIG. 22A, the sheath region **730** indicates walls that are suctioned and collapsed toward one another separated only by the interior frame **726**. As can be seen in FIG. 22A, the expandable dielectric sheath **725** and frame **726** is properly expanded within the uterine cavity **302** and no perforations are shown. In this case, referring back to FIGS. 15 and 16 and the accompanying text, the cavity integrity test would provide an inflow of CO.sub.2 at an

initial or first flow rate of at least 0.040 slpm (if measured in free air). The CO.sub.2 inflow would continue for up to 30 seconds or until the flow rate decayed and remained below a second flow rate for a selected time interval of at least 1 second, at least 2 seconds, 5 at least seconds, or at least 10 seconds. In one system and method, the second flow rate is less than the first flow rate and less than 0.050 slpm. In one method, the initial or first flow rate (in free air) is 0.070 slpm, the inflow is continued for 30 seconds or until the flow rate decays to 0.034 slpm and then the flow rate continuously remains below 0.034 slpm for 5 seconds. If the previous conditions are met, then the controller **710** would display a message and signal that the test indicates that the uterine cavity is non-perforated. In one embodiment, the controller **710** and operational algorithm is configured to automatically activate the RF source to thereby initiate the endometrial ablation treatment. In another embodiment, the controller **710** is configured only to enable the RF source and thereafter the physician can manually activate the RF source to initiate the ablation procedure.

(84) The utility of the above-described cavity integrity test can be understood with reference to FIG. **22B**, which depicts the working end **700** and dielectric sheath **725** within an exemplary perforation **736** in the uterine wall tissue **738**. In FIG. **22B**, it can be seen that the sheath **715** and interior frame **726** have a partially expanded configuration and extending through a perforation that could have been created by a 'sound' instrument used to measure dimensions of the uterine cavity. FIG. **22B** depicts the controller **715** and algorithm after actuation of the cavity integrity test wherein negative pressure source **720** is actuated to suction gas from interior chamber **722** of dielectric sheath **725**. In FIGS. **22B** and **23**, it can be seen that thin-wall dielectric sheath **725** is under negative with sheath wall regions **730'** suctioned and collapsed toward one another and separated only by interior frame **726**. FIG. **23** is an enlarged cut-away view of the sheath **725** in the perforation of FIG. **22B** and shows best how the suctioned down sheath region **730'** leaves a trough **740** between tissue **738** (hatched region) and the sheath **725** through which inflowing CO.sub.2 can escape from the uterine cavity **302** through the perforation **736** in tissue **738**. Under the previous test parameters, the CO.sub.2 flow would escape the uterine cavity **302** and the test algorithm would find that there was no flow decay over a selected time interval (e.g., 5 to 30 seconds). Thus, the controller **715** would display a message that a perforation existed and the RF source would be disabled. In contrast, if the sheath **725** was not under negative pressure, it can be understood that the sheath **725** and interior frame **726** could plug the perforation and thus prevent CO.sub.2 escape—which would then result in flow decay which in turn would mask the perforation **736**.

(85) The cavity integrity test described above with reference to FIGS. **22A-22B** can be used as a single stage test or it can be used sequentially with the earlier described test of FIGS. **15-16**. Such a two-stage test could add an additional level of safety to cavity integrity testing.

(86) FIG. **24** illustrates a two stage cavity integrity test wherein the first stage is similar to that of FIGS. **15-16** and the second stage is the test described with reference to FIGS. **22A-22B**. More in particular, gas inflow into the interior chamber **722** of the dielectric **725** is provided until the pressure reaches a predetermined level which in one algorithm is 0.50 psi. Thereafter, CO.sub.2 inflows into the uterine cavity are initiated as described above and then flow decay is monitored until flow diminished to less than a predetermined level which in one algorithm is 0.034 slpm. If this first aspect of the test is achieved in less than 10 seconds, 30 seconds or 60 seconds, then the second stage of the test is as described with reference to FIGS. **22A-23** wherein the interior chamber **722** of the dielectric **725** is suctioned down against frame **726** and the flow decay test is repeated over another predetermined time interval which can be from 10 to 60 seconds. In all other respects, the two-stages test depicted in FIG. **24** corresponds to the stages described individually above.

(87) Although particular embodiments of the present invention have been described above in detail, it will be understood that this description is merely for purposes of illustration and the above description of the invention is not exhaustive. Specific features of the invention are shown in some drawings and not in others, and this is for convenience only and any feature may be combined with

another in accordance with the invention. A number of variations and alternatives will be apparent to one having ordinary skills in the art. Such alternatives and variations are intended to be included within the scope of the claims. Particular features that are presented in dependent claims can be combined and fall within the scope of the invention. The invention also encompasses embodiments as if dependent claims were alternatively written in a multiple dependent claim format with reference to other independent claims.

(88) Other variations are within the spirit of the present invention. Thus, while the invention is susceptible to various modifications and alternative constructions, certain illustrated embodiments thereof are shown in the drawings and have been described above in detail. It should be understood, however, that there is no intention to limit the invention to the specific form or forms disclosed, but on the contrary, the intention is to cover all modifications, alternative constructions, and equivalents falling within the spirit and scope of the invention, as defined in the appended claims.

(89) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. The term “connected” is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

(90) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

(91) All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

Claims

1. A method of evaluating an integrity of a patient's uterus, the method comprising: positioning an expandable structure in a uterine cavity of a patient, the expandable structure comprising a portion of an energy application system for delivering ablation energy to uterine tissue; expanding the expandable structure within the uterine cavity; introducing a fluid from a first fluid source into the uterine cavity exterior of the expandable structure; introducing a fluid from a second fluid source into the expandable structure; monitoring a flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure; and monitoring a flow rate of the fluid introduced into the

expandable structure; determining whether the uterine cavity is perforated or non-perforated, wherein the uterine cavity is determined as perforated where the flow rate of the fluid introduced into the expandable structure impinges the flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure.

2. The method of claim 1, further comprising: permitting activation of RF energy to the energy application system based on a determination the uterine cavity is non-perforated.

3. The method of claim 1, further comprising: preventing activation of RF energy to the energy application system based on a determination the uterine cavity is perforated.

4. The method of claim 1, wherein the flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure and the flow rate of the fluid introduced into the expandable structure are monitored contemporaneously.

5. The method of claim 1, wherein the flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure and the flow rate of the fluid introduced into the expandable structure are monitored sequentially.

6. The method of claim 1, wherein introducing the fluid from the first fluid source into the uterine cavity exterior of the expandable structure is initiated prior to introducing the fluid from the second fluid source into the expandable structure.

7. The method of claim 1, wherein the fluid from the first fluid source is CO.sub.2.

8. The method of claim 1, wherein the fluid from the second fluid source is Argon.

9. The method of claim 1, wherein the fluid from the second fluid source is introduced into the expandable structure at a flow rate of 0.8 SLPM.

10. The method of claim 1, wherein expanding the expandable structure within the uterine cavity includes expanding a frame within an interior of the expandable structure.

11. A method of evaluating an integrity of a patient's uterus, the method comprising: positioning an expandable structure in a uterine cavity of the patient, the expandable structure comprising a portion of an energy application system for delivering ablation energy to uterine tissue; expanding the expandable structure within the uterine cavity; introducing a fluid from a first fluid source into the uterine cavity exterior of the expandable structure; introducing a fluid from a second fluid source into the expandable structure; and determining whether the uterine cavity is perforated or non-perforated based on a monitored flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure and/or a monitored flow rate of the fluid introduced into the expandable structure; wherein the uterine cavity is determined as perforated where the flow rate of the fluid introduced into the expandable structure impinges the flow rate of the fluid introduced into the uterine cavity exterior of the expandable structure; and wherein the flow rate of the fluid introduced into the uterine cavity is impinged where the flow rate of the fluid introduced into the uterine cavity changes during the introduction of the fluid into the expandable structure.

12. The method of claim 11, wherein the fluid from the first fluid source is CO.sub.2.

13. The method of claim 11, wherein the fluid from the second fluid source is Argon.

14. The method of claim 11, further comprising: permitting activation of RF energy to the energy application system based on a determination the uterine cavity is non-perforated.

15. The method of claim 11, further comprising: preventing activation of RF energy to the energy application system based on a determination the uterine cavity is perforated.

16. The method of claim 11, wherein introducing the fluid from the first fluid source into the uterine cavity exterior of the expandable structure is initiated prior to introducing the fluid from the second fluid source into the expandable structure.

17. A method of evaluating an integrity of a patient's uterus, the method comprising: positioning an expandable structure in a uterine cavity of the patient, the expandable structure comprising a portion of an energy application system for delivering ablation energy to uterine tissue; expanding the expandable structure within the uterine cavity; introducing a first gas into the uterine cavity exterior of the expandable structure; introducing a second gas into an interior of the expandable

structure; monitoring a flow rate of an inflow of the first gas with a flow meter; determining whether the flow rate of the first gas changes during an inflow of the second gas into the interior of the expandable structure; and characterizing the uterine cavity as perforated based on a determination the flow rate of the second gas introduced into the expandable structure impinges the flow rate of the first gas introduced into the uterine cavity exterior of the expandable structure.

18. The method of claim 17, further comprising: preventing activation of RF energy to the energy application system based on a determination the uterine cavity is perforated.

19. The method of claim 17, wherein introducing the second gas into the interior of the expandable structure is initiated subsequent to introducing the first gas into the uterine cavity exterior of the expandable structure.

20. The method of claim 17, wherein the second gas is introduced into the expandable structure at a flow rate of 0.8 SLPM.
